

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 01 • Proof

Questions

Student book • questions with writing space

- 31 pages
- Past papers up to January 2026

PURE 2 | TOPIC 01

Proof

Your classified route

15

QUESTIONS

75

MARKS

4

FAMILIES

5

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 01 | FAMILY 01

Exhaustion/Cases

WHAT THIS FAMILY ASKS

Recognise the exhaustion/cases demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

V02 For modulo cases, use $3k, 3k+1, 3k+2$ or $4k, 4k+1, 4k+2, 4k+3$.

8 classified questions

36 total marks

10. (i) Prove by counter example that the statement

“if p is a prime number then $2p + 1$ is also a prime number”

is not true.

(1)

Adaptive unit 1 · WMA12-01_2022_Jan_Q10

DR ESAM AHMED

(ii) Use proof by exhaustion to prove that if n is an integer then

$$5n^2 + n + 12$$

is always even.

(4)

Adaptive unit 2 · WMA12-01_2022_Jan_Q10

DR ESAM AHMED

1. Given that a, b and c are integers greater than 0 such that
- $c = b + 2$
 - $a + b + c = 10$

Prove, by exhaustion, that the product of a, b and c is always even.

You may use the table below to illustrate your answer.

(3)

Adaptive unit 1 · WMA12-01_2022_Oct_Q01

DR ESAM AHMED

8. (i) A student writes the following statement:
“When a and b are consecutive **prime** numbers, $a^2 + b^2$ is never a multiple of 10”
Prove by counter example that this statement is **not** true.

(2)

Adaptive unit 1 · WMA12-01_2023_MayJune_Q08

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(ii) Given that x and y are even integers greater than 0 and less than 6, prove by exhaustion, that

$$1 < x^2 - \frac{xy}{4} < 15$$

(3)

Adaptive unit 2 · WMA12-01_2023_MayJune_Q08

DR ESAM AHMED

1. Given that a, b and c are integers greater than 0 such that
- $c = 3a + 1$
 - $a + b + c = 15$

prove, by exhaustion, that the product abc is always a multiple of 4

You may use the table below to illustrate your answer.

(3)

Adaptive unit 1 · WMA12-01_2023_Oct_Q01

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8. (i) Use a counter example to show that the following statement is **false**

$$“n^2 + 3n + 1 \text{ is prime for all } n \in \mathbb{N}”$$

(2)

Adaptive unit 1 · WMA12-01_2024_Jan_Q08

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(ii) Use algebra to prove by exhaustion that for all $n \in \mathbb{N}$

“ $n^2 - 2$ is **not** a multiple of 4”

(4)

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DR ESAM AHMED

11. (i) Prove by counter example that the statement

“If n is a prime number then $3^n + 2$ is also a prime number.”

is false.

(2)

Adaptive unit 1 · WMA12-01_2024_Oct_Q11

DR ESAM AHMED

(ii) Use proof by exhaustion to prove that if m is an integer that is **not** divisible by 3, then

$$m^2 - 1$$

is divisible by 3

(4)

Adaptive unit 2 · WMA12-01_2024_Oct_Q11

DR ESAM AHMED

10: In this question you must show detailed reasoning.

Use algebra to prove by exhaustion that,

for all positive integers m that are **not** multiples of 3, the value of

$$m^2 + 3m + 2$$

is always a multiple of 3

(4)

Adaptive unit 1 · WMA12-01_2025_MayJune_Q10

DR ESAM AHMED

10. A student was asked to prove by exhaustion that

if n is an integer then $2n^2 + n + 1$ is **not** divisible by 3

The start of the student's proof is shown in the box below.

Consider the case when $n = 3k$

$$2n^2 + n + 1 = 18k^2 + 3k + 1 = 3(6k^2 + k) + 1$$

which is not divisible by 3

Complete this proof.

(4)

Adaptive unit 1 · WMA12-01_2023_Jan_Q10

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TOPIC 01 | FAMILY 02

Disproof

WHAT THIS FAMILY ASKS

Recognise the disproof demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**2 classified questions**

13 total marks

3. (i) Show that the following statement is **false**:

$$“(n + 1)^3 - n^3 \text{ is prime for all } n \in \mathbb{N}”$$

(2)

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DR ESAM AHMED

(ii) Given that the points $A(1, 0)$, $B(3, -10)$ and $C(7, -6)$ lie on a circle, prove that AB is a diameter of this circle.

(5)

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5.

In this question you must show detailed reasoning.(a) Given that x and y are positive numbers such that

$$(x - y)^3 > x^3 - y^3$$

prove that

$$y > x$$

(4)

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5. In this question you must show detailed reasoning.

(b) Using a counter example, show that the result in part (a) is not true for all real numbers.

(2)

Adaptive unit 2 · WMA12-01_2024_MayJune_Q05

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TOPIC 01 | FAMILY 03

Algebraic Divisibility

WHAT THIS FAMILY ASKS

Recognise the algebraic divisibility demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**3 classified questions**

18 total marks

- (2)

(ii) Show, using algebra, that for $n \in \mathbb{N}$

$$(n + 1)^3 - n^3 \text{ is not a multiple of } 3$$

(3)

Adaptive unit 2 · WMA12-01_2021_MayJune_Q03

DR ESAM AHMED

8. (i) A student states

“If x and y are irrational numbers, $x \neq y$, then xy is also irrational.”

Show, by counter example, that this statement is not always true.

(1)

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(ii) Prove, using algebra, that for all odd integers n , the value of the expression

$$n^3 + 3n + 2$$

is always even but never a multiple of 4

(4)

Adaptive unit 2 · WMA12-01_2025_Jan_Q08

DR ESAM AHMED

6. (i) Given that p and q are consecutive odd numbers, where $p > q > 0$, prove that

$$p^2 - q^2$$

is a multiple of 8

(4)

Adaptive unit 1 · WMA12-01_2026_Jan_Q06

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(ii) The curve C has equation

$$y = x^3 + 12x^2 + 49x + 2$$

Prove that C has no stationary points.

(4)

Adaptive unit 2 · WMA12-01_2026_Jan_Q06

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TOPIC 01 | FAMILY 04

Inequality Proof

WHAT THIS FAMILY ASKS

Recognise the inequality proof demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

2 classified questions

8 total marks

5. (i) Use algebra to prove that for all $x \geq 0$

$$3x + 1 \geq 2\sqrt{3x}$$

(3)

Adaptive unit 1 · WMA12-01_2021_Jan_Q05

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(ii) Show that the following statement is not true.

“The sum of three consecutive prime numbers is always a multiple of 5”

(1)

Adaptive unit 2 · WMA12-01_2021_Jan_Q05

DR ESAM AHMED

9. (a) Prove that for all positive values of x and y ,

$$\frac{x + y}{2} \geq \sqrt{xy}$$

(3)

Adaptive unit 1 · WMA12-01_2021_Oct_Q09

DR ESAM AHMED

(b) Prove by counter-example that this inequality does not hold when x and y are both negative.

(1)

Adaptive unit 2 · WMA12-01_2021_Oct_Q09

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